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Project Overview:

This project builds an AI system that understands fashion products from images and lets users search, compare, and organize them intelligently — without ever needing manually written product tags.

In simple terms, the system looks at a clothing photo (like a shirt or a pair of shoes) and converts it into a list of numbers called an 'embedding'. Similar-looking products end up with similar numbers, which is what makes searching and comparing them possible.

Three features were built on top of this foundation:

- **Smart Product Recommendation Engine** — suggests items that go well with a product (e.g., socks and a fitness watch for running shoes).
- **Unique Product Catalog Creation** — automatically detects near-duplicate product photos and keeps only one representative item per group.
- **Reverse Product Search** — lets a user type a plain-text description (e.g., 'blue casual shirt') and get back the closest matching products.

Dataset: <https://www.kaggle.com/datasets/paramaggarwal/fashion-product-images-small>

Overall Architecture:

All three features share one core part, turn images and text into comparable number vectors using a model CLIP, then use fast similarity search to find related images.

Step1: Building the Image Embedding: Every product photo is passed through CLIP model which converted into vector. Visually similar images have close number's vector.

Approch: Dataset Img -> CLIP Vision Encoder -> Image Embedding -> FAISS Index

Step2: FAISS: All 15,596 embeddings are stored in FAISS Index. FAISS finds the closest matching products.

Task1 – Smart Product Recommendation Engine:

Approach: Rather than guessing complementary items from raw pixels, the notebook uses simple, transparent rule-based mapping from product category to complementary categories that makes easy to read, extend, and explain.

Input Product -> Product Category -> Recommendation Rules -> Complementary Products

Product Category	Recommended Categories
Shoes	Socks, Fitness Watch, Water Bottle
Shirts	Jeans, Belt, Sunglasses
Tshirts	Jeans, Sneakers, Cap

Result:

```
recommend("Shoes")

['Socks', 'Fitness Watch', 'Water Bottle']
```

Task 2 – Unique Product Catalog Creation:

Approach: Using the same embedding from Step 1, the notebook applies DBSCAN clustering, which automatically groups visually similar products together without needing to be told how many group to be formed. Or it also easily separate the anomaly.

Product Image -> CLIP Embedding -> DBSCAN -> Duplicate Groups -> Unique Catalog

Result:

Total Products: 15596
Unique Catalogs: 45
Duplicates Removed: 15551

Catalog_ID		Catalog_Name
0	0	Turtle Check Men Navy Blue Shirt
1	1	Nike Unisex U Black Caps
2	2	FIFA Mens Essentials Red T-shirt
3	3	Reebok Kids Kkr Boy's Premium Blue Jerseys
4	4	ADIDAS Women Pink Socks
5	5	Nike Unisex Casual Grey Backpack
6	6	ADIDAS Unisex NBA Black Cap
7	7	Nike Women As See The Win Black T-Shirts
8	8	ADIDAS Men Green South Africa Cricket Jersey
9	9	Puma Unisex Liga XL Football

Note: with the distance setting used (eps = 0.15), DBSCAN formed 45 broad style clusters (e.g., all blue checked shirts in one group) rather than tight exact-duplicate groups. A smaller eps value would produce tighter clusters closer to true near-duplicate detection — a good next tuning step.

Task 3 – Reverse Product Search (Text-to-Image)

Approach: CLIP understand both images and texts in the same shared number space. A type sentence can be converted into a vectors and its directly compare with image which is already stored in FAISS DB.

Text Query -> CLIP Text Encoder -> Text Embedding -> FAISS Search -> Top-K Similarity products.

Result:

query = "blue casual shirt"

Top-5 Similarity returns:

1. Wrangler Men Multi Blue Shirt | Score: 0.2937
2. Peter England Men Check Navy Blue Shirt | Score: 0.2917
3. Highlander Men Check Blue Shirt | Score: 0.2915
4. U.S. Polo Assn. Men Solid Blue Shirt | Score: 0.2908
5. Scullers For Her Check Blue Shirt | Score: 0.2904

The Images are appear:

